FERNANDO A. ORTEGA

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EDUCATION

Baylor College of Medicine — **Houston, TX**
Ph.D. in Neuroscience — **Aug 2024 - Present**

St. Olaf College — **Northfield, MN**
BA in Biology and Psychology — **Sept 2018 - May 2022**

RESEARCH

University of Minnesota Department of Neuroscience — **Minneapolis, MN**
Researcher at Heilbronner Lab — **June 2022 - Present**

- Skillfully employing cutting-edge MRI diffusion tractography techniques to elucidate the complex connectivity patterns of the posteromedial cortex.
- Contributing to bridging the knowledge gap between in vivo imaging and ex vivo tract-tracing methodologies, thereby enhancing our understanding of the human brain's mesoscale anatomical structure.

St. Olaf College Department of Biology and Neuroscience Program — **Northfield, MN**
Undergraduate Researcher at Lee Lab — **May 2021 - May 2022**

- Contributed to experimental data collection investigating auditory processing and behavioral responses of the parasitoid fly *Ormia ochracea* to evolving acoustic signals produced by host crickets.
- Investigated the behavioral changes in *Ormia ochracea* when introduced to alcohol in their food source and provided insights into the impact of substances on auditory and navigational behaviors.
- Utilized DeepLabCut technology for precise animal pose estimation, providing accurate tracking and analysis of fly behavior.

St. Olaf College Department of Psychology — **Northfield, MN**
Undergraduate Researcher at Petok Lab — **Sept 2021 - Dec 2021**

- Conducted an in-depth examination of the influence of fMRI and structural MRI presentations on the perceived validity of scientific articles, enhancing understanding of how neuroimaging data presentation affects public perception.
- Engaged in all aspects of a comprehensive semester-long research project, from initial literature review, study design and crafting of the IRB proposal, through to hands-on data collection and sophisticated data analysis using SPSS.

University of Minnesota Department of Plant Pathology — **Northfield, MN**
Summer Undergraduate Student Researcher at Dill-Macky Lab — **June 2019 - Sept 2019**

- Conducted research on plant pathogens, honing laboratory techniques such as DNA extraction and loop-mediated isothermal amplification (LAMP) assays, foundational experiences that can be applied to neuroscientific studies.
- Acquired expertise in the use of sophisticated analytical methodologies, specifically Duveiller's (Duveiller, 1992) method, for the assessment and quantification of disease severity, equipping me with analytical skills readily applicable to the interpretation and evaluation of neuroimaging data.

PROFESSIONAL DEVELOPMENT

Minnesota Inclusive Neuroscience Development Scholars (MINDS) Program — **Minneapolis, MN**
Two year post-baccalaureate program in neuroscience — **July 2022 - Present**

- Receiving individualized mentorship and professional skill development with the goal of becoming a future leader in neuroscience.

University of Washington eScience Institute **Seattle, Washington**
Neurohackademy - Intensive MRI summer course **August 7-18, 2023**

- Developed expertise in using various technologies for the analysis of human neuroscience data, with a focus on the shareability and reproducibility of results.

DIPY Workshop 2023 **Virtual**
Diffusion Imaging in Python program **April 24-28, 2023**

- Engaged in a comprehensive online workshop, designed to sharpen skills and deepen knowledge in the realm of python in structural and diffusion imaging.

PUBLICATIONS

- **Ortega, F. A**., Hayday, E., & Heilbronner, S. R. (2026). Fiber pathways of the posterior cingulate cortex. *Cerebral Cortex*. Manuscript under review (second revision submitted October 27, 2026).
- Tang, W., Monko, M. E., Liu, Z., Manea, A. M. G., **Ortega, F. A**., Hart, D., Zhou, J., Zimmermann, J., & Heilbronner, S. R. (2026). Functional vs anatomical cortico-striatal connectivity in the macaque brain. *Translational Psychiatry, 16(1), Article 13.* https://doi.org/10.1038/s41398-025-03757-x
- Jafri, R., **Ortega, F. A**., Manivannan, P., Jourahmad, Z., Devara, D., et al. (2026). Electrophysiological features of signals recorded from white matter. *bioRxiv* [Preprint]. https://doi.org/10.64898/2026.07.11.737939
- DiMaggio-Potter ME, Velosa A, Brent IV JC, Del Rio O, Abebaw EF, **Ortega F. A,** et al. (2025) Ten simple rules for thriving in a post-baccalaureate research program. PLoS Comput Biol 21(7): e1013174. https://doi.org/10.1371/journal.pcbi.1013174

POSTER PRESENTATIONS & RESEARCH CONTRIBUTION

- Wikle, A. W., Broder, E. D., Gallagher, J. H., Dominguez, J., Carlson, M., Vu, Q., Tinghitella, R. M., & Lee, N. (2025). Neural and behavioral evolution in an eavesdropper with a rapidly evolving host. Current Biology, 35(5), 1074–1084.e7.
- **Ortega, F. A**., Bullock, D. N., Liu, Z., Zimmermann, J., & Heilbronner, S. R. "*Structural Connectivity of the Posteromedial Cortex*." Poster session presented at the Annual Biomedical Research Conference for Minoritized Scientists, Phoenix, AZ: November 15, 2023.
- **Ortega, F. A**., Bullock, D. N., Liu, Z., Zimmermann, J., & Heilbronner, S. R. "*Structural Connectivity of the Posteromedial Cortex*." Poster session presented at the Society for Neuroscience, Washington, DC: November 9, 2023.
- **Ortega, F. A**., Bullock, D. N., Liu, Z., Zimmermann, J., & Heilbronner, S. R. "*Structural Connectivity of the Posteromedial Cortex*." Poster session presented at the University of Minnesota Department of Neuroscience Postbaccalaureate Poster Symposium, Minneapolis, MN: June 18, 2023.
- **Ortega, F. A.**, Aarrestad, I., Dickinson, S., Lee, N. "*Quantifying Behavioral Preference For Ethanol of The Parasitoid Fly Ormia ochracea (Diptera: Tachinidae).*" Poster session presented at St. Olaf College Honors Day Poster Symposium: May 2022
- **Ortega, F. A**., Curland, R., & Dill-Macky, R. "*Bacterial Leaf Streak of Wheat: How Identifying Resistant Varieties and Rapidly Detecting the Pathogen Contribute to Managing the Disease.*" Poster session presented at the Summer Undergraduate Research Expo, McNamara Alumni Center, Memorial Hall: August 7, 2019

TEACHING EXPERIENCE

Go4Brains **Minneapolis, MN**
High-School Neuroscience Summer Program **July, 2022**

- Mentored a small group of high school students about how to apply to college and how to find neuroscience research experiences.

Fifth Day Experience **Coon Rapids, MN**
Paladin High School **May, 2022**

- Taught a class of high school students about the neuroscience of memory and amnesia.

Mysteries of Your Brain at Bell Museum **Minneapolis, MN**

Invited to present **April, 2022**

- Engaged with visitors, providing educational insights into the structural anatomy of the brain, enriching their understanding of its complex structure and function.

St. Olaf College Biology Department **Northfield, MN**

Genetics Lab Teaching Assistant (TA) **Feb 2021 - May 2021**

- Assisted the instructor in implementing the lab curriculum, ensuring a coherent and effective learning experience.

ACTIVITIES & LEADERSHIP

Co- Chair of Professional Development Committee at MINDS **Minneapolis, MN**

Co-Chair **July 2022 - Present**

- As a Co-Chair of the Professional Development Committee at MINDS, I played an important role in designing and implementing initiatives to advance the professional skills of our members.

Community Engagement at MINDS **Minneapolis, MN**

Member ***July 2022 - Present***

- Contributed to outreach programs aimed at bringing neuroscience awareness to the broader community. I participated in organizing public presentations, school visits, and interactive activities that made neuroscience accessible and engaging to people of all ages.

Social Committee at MINDS. **Minneapolis, MN**

Member ***July 2022 - Present***

- Involved in creating and coordinating social events that promoted a sense of camaraderie and wellbeing among MINDS members.

MINDS Journal Club **Minneapolis, MN**

Member ***July 2022 - Present***

- Participating as a Member of the MINDS Journal Club, I engaged in critical analysis and discussion of cutting-edge neuroscience research with fellow scholars. This platform allowed me to stay abreast of the latest developments in the field and refine my analytical thinking.

The Premier Undergraduate Neuroscience Journal **Northfield, MN**

Peer Reviewer **Sept 2021 - May 2022**

- Meticulously evaluated research manuscripts, providing constructive feedback to improve the quality and integrity of the submissions.

SKILLS

- Advanced use of FSL for brain imaging data analysis.
- Skilled in diffusion magnetic resonance imaging (dMRI) and tractography for neural connectivity studies. Knowledgeable in MRtrix, DIPY, TractSeg, pyAFQ.
- Proficient in Python scripts for MRI statistical analysis and visualization.
- Experienced using High Performance Computing (HPC) systems
- Skilled with IMOD for 3D visualization and analysis of biological microscopy data.
- Knowledgeable in using DeepLabCut for 3D markerless pose estimation with high accuracy.
- Knowledgeable about retro and anterograde tract tracing.
- Experienced with injecting tract tracers into macaque brains

AWARDS AND HONORS

- **NIH T32 Predoctoral Training Fellowship,** Baylor College of Medicine, 2024
- **Full Travel Award,** Annual Biomedical Research Conference for Minoritized Scientists, 2023
- **Poster Award,** University of Minnesota Department of Neuroscience Postbaccalaureate Symposium, 2023.
- **Nu Rho Psi**, National Honor Society in Neuroscience, 2022
- **Adam S. Thomas Award,** funding for one year long research project at St. Olaf College, 2022
- **Dean's List (3x),** received a 3.75 GPA or higher in a semester
- **TRIO McNair Scholars Program,** St Olaf College 2020-2022